

Jinwoo Chung

✉ imaboybut@kookmin.ac.kr

RESEARCH INTERESTS

Efficient and Trustworthy Foundation Models, Quantization (PTQ/QAT), Model Compression, VLMs, Diffusion Models, KV-Cache Quantization, Low-Bit Deployment.

EDUCATION

Kookmin University, Korea

Aug. 2025 – Feb. 2027 (expected)

M.S. in *Computer Engineering* (GPA: 4.31/4.5)

- Adviser: Prof. Jangho Kim

Kookmin University, Korea

Mar. 2022 – Aug. 2025

B.S. in *Artificial Intelligence* (GPA: 4.0/4.5)

CURRENT RESEARCH

Behavior-Aware Quantization for Foundation Models

2026 – Present

- Studying how low-bit quantization changes generative behavior in diffusion models, especially minority generation and sample diversity.
- Analyzing how activation quantization alters attention patterns and cross-modal routing in vision-language models.
- Exploring behavior-preserving quantization methods for efficient and reliable foundation model deployment.

PUBLICATIONS

<International Conferences>

- [C5] **Hardware-Friendly Token-Group Activation Quantization for Low-Bit Mamba Super-Resolution**

Jinwoo Chung, Jangho Kim

Neural Information Processing Systems (**NeurIPS**), 2026

- [C4] **Temporal-Aware Quantization via Sequence-Wise Quantizer Bounds Ensembling for Real-World Video Super-Resolution**

Jinwoo Chung, Sangho An, Sungyeop Jung, Jangho Kim

European Conference on Computer Vision (**ECCV**), 2026

- [C3] **SR-BEVS: Sparse Recovery Bird’s-Eye View Model Soup for Robust Camera-Based 3D Object Detection**

Jinwoo Chung, Sangho An, Hyunsung Park, Jangho Kim

IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2026

- [C2] **Quantized Model Soup Shake-Up: Weight Perturbation for Enhanced Ensemble Diversity**

Jinwoo Chung, Sungyeop Jung, Weronika Czorapinska, Jangho Kim

ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2026

- [C1] **SQuaT: Self-Supervised Knowledge Distillation via Student-Aware Quantized Teacher Features**
HyeonJun Lee*, Hyeonsik Jo*, **Jinwoo Chung**, Jangho Kim
International Conference on Artificial Intelligence and Statistics (**AISTATS**), 2026

<Under Review>

- [U2] **Gradient Interaction Sensitivity for Structured Pruning of Large Language Models**
Sangho An*, **Jinwoo Chung***, Jangho Kim
Under review

- [U1] **Quantized Audio Spectrogram Transformer Soup: Robustness and Deployment Efficiency**
Jinwoo Chung, Songha Lee, Jangho Kim
Short paper, under review

* Equal contribution

HONORS & AWARDS

Outstanding Paper Award, ICROS 2026 2026
Awarded for “Reinforcement Learning-Based Quantization Scale Optimization for Low-Bit Vision Transformers”

PROJECTS

On-Device AI Quantization Algorithm Development Jan. 2025 – Dec. 2025
Industry-Academia Project (LG Electronics)

- Developed quantization algorithms for on-device AI deployment.
- Submitted a research paper proposing three quantization methods.

Autonomous Driving Perception Robustness Jan. 2025 – Dec. 2025
Industry-Academia Project (Hyundai Motor Company)

- Built a road-icing segmentation dataset for low-temperature driving scenarios.
- Developed a robustness-enhancing method for camera-based BEV 3D object detection.